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**ROBUST CONTROL OF LINEAR SYSTEMS WITH
SWITCHED ACTUATORS SUBJECTED TO
DWELL-TIME CONSTRAINTS**

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ROBUST CONTROL OF LINEAR SYSTEMS WITH SWITCHED ACTUATORS SUBJECTED TO DWELL-TIME CONSTRAINTS

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*"I am and always will be the optimist.
The hoper of far-flung hopes, and the dreamer of improbable dreams."*
— THE ELEVENTH DOCTOR

Resumo

Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o sistema à trajetória cíclica alvo.

Abstract

Switched actuators are difficult to handle because of some characteristic constraints of their nature. but they are efficient in numerous engineering applications, being applied, for example, in attitude control systems for space vehicles. Thus, knowing how to use this type of actuator is important and necessary. The main problem to be solved in this thesis is the design of a controller that guarantees the robustness of a system with switched actuators subjected to time constraints that limit the response window of their commands. In this approach, a limit cycle target trajectory search is performed and then the error of the states compared to the target trajectory is calculated. Then a predictive control technique is applied to correct the error and get the system to the cyclic target trajectory

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List of Abbreviations and Acronyms

CTq	computed torque
DC	direct current
EAR	Equação Algébrica de Riccati
GDL	graus de liberdade
ISR	interrupção de serviço e rotina
LMI	linear matrices inequalities
MIMO	multiple input multiple output
PD	proporcional derivativo
PID	proporcional integrativo derivativo
PTP	point to point
UARMII	Underactuated Robot Manipulator II
VSC	variable structure control

List of Symbols

a	Distância
$\mathbf{a}$	Vetor de distâncias
$\mathbf{e}_j$	Vetor unitário de dimensão n e com o j -ésimo componente igual a 1
$\mathbf{K}$	Matriz de rigidez
m_1	Massa do cumpim
δ_{k-k_f}	Delta de Kronecker no instante k_f

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1 Introduction

1.1 Motivation

Switched control system is used in a variety of situations in engineering (space, electrical, ...) and it is importante to understand and master its use for a better ...

1.2 Contributions of the present work

This work addresses a ongoing line of research where the main goal is to control systems with actuators subjected to switching-time constraints. This follows the work of Vieira (2013), where the control actions are restricted to integer values in the set $\{-1, 0, 1\}$ and Marcolino (2018), where the control actions are applied as perturbations on the switching times to reach a periodic target trajectory calculated as a optimization problem.

The work so far considers the system as a invariant system, but this does not represent all cases in this class of problem. Therefore this work proposes a time-varying system approach, thus being a more general case of the solution for predictive control of switched systems.

1.3 Outline of the remaining chapters

O capítulo 1 contém ...

2 Determination of Target Periodic Trajectories

2.1 Statement of the problem

2.2 Proposed codification for the switching constraints

2.3 Optimization

description of the optimization for finding the target trajectory

2.4 Calculating x_r

$$x_0 \in \mathfrak{R}$$

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau \quad (2.1)$$

Modos:

$$\begin{aligned} [0 - t_1] : \dot{x} &= A_1x + B_1u_1 \\ [t_1 - t_2] : \dot{x} &= A_2x + B_2u_2 \\ &\vdots \\ [t_{N-1} - t_N] : \dot{x} &= A_Nx + B_Nu_N \end{aligned} \quad (2.2)$$

$$\begin{aligned}
x(t_1) &= e^{A_1 t_1} x(0) + \int_0^{t_1} e^{A_1(t_1-\tau)} B_1 u_1 d\tau \\
x(t_1) &= e^{A_1 t_1} x(0) + \left[\int_0^{t_1} e^{A_1(t_1-\tau)} B_1 d\tau \right] u_1 \\
x(t_1) &= F_1 x(0) + G_1 u_1
\end{aligned} \tag{2.3}$$

$$x(t_1) = F_1 x(0) + G_1 u_1 \tag{2.4}$$

$$x(t_2) = F_2 x(1) + G_2 u_2 \tag{2.5}$$

$$\vdots$$

$$x(t_N) = F_N x(N-1) + G_N u_N \tag{2.6}$$

$$\begin{aligned}
x(t_N) &= F_N F_{N-1} \dots F_1 x(0) + F_N F_{N-1} \dots F_2 G_1 x(1) + \\
&\quad F_N F_{N-1} \dots F_2 G_2 x(2) + \dots + F_N G_{N-1} x(N-1) + G_N x(N) \\
x(t_N) &= F_N F_{N-1} \dots F_1 x(0) + c
\end{aligned} \tag{2.7}$$

Queremos $x(t_N) = x(0) = ?$

$$\begin{aligned}
x(0) &= F_N F_{N-1} \dots F_1 x(0) + c \\
\underbrace{(I - F_N F_{N-1} \dots F_1)}_{\text{suposta invertível}} x(0) &= c \\
x(0) &= (I - F_N F_{N-1} \dots F_1)^{-1} c
\end{aligned} \tag{2.8}$$

em que

$$\begin{aligned}
c &= F_N F_{N-1} \dots F_2 G_1 x(1) + \\
&\quad F_N F_{N-1} \dots F_2 G_2 x(2) + \dots + F_N G_{N-1} x(N-1) + G_N x(N)
\end{aligned} \tag{2.9}$$

2.5 Application Examples

2.5.1 Double Integrator

2.5.2 Variant Dynamic Circuit 1

2.5.3 Variant Dynamic Circuit 2

3 Predictive Control for Stabilization of Periodic Trajectories

3.1 Statement of the problem

Statement of the problem

Consider a plant with continuous-time dynamics described by a model of the form

$$\dot{x}(t) = A_c x(t) + B_c u(t) \quad (3.1)$$

3.2 Linearization procedure

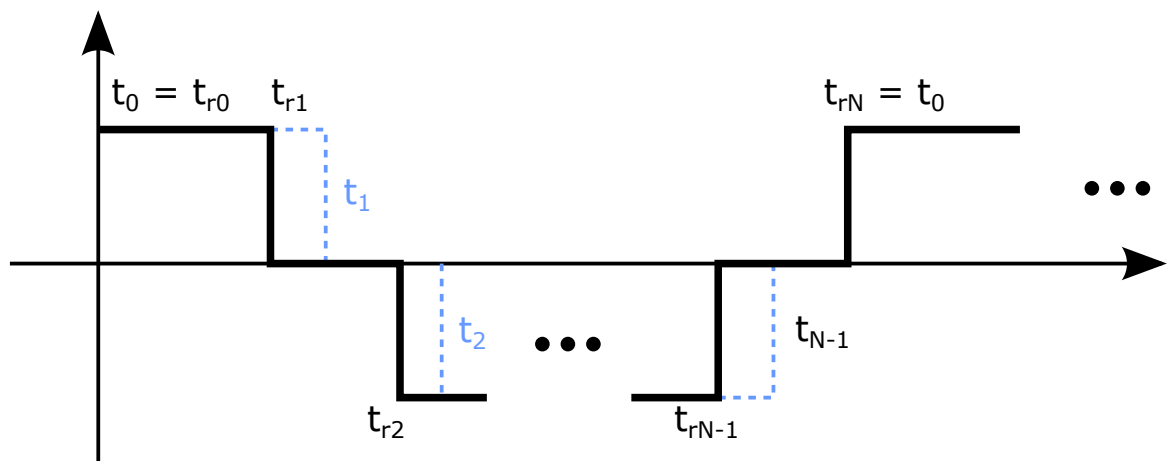


FIGURE 3.1 – Instantes de chaveamento

The solution for the state equation (3.1) at time t_N , starting from the initial condition $x(t_0)$, is of the form

$$x(t_1) = e^{A_0(t_1-t_0)}x(t_0) + \int_{t_0}^{t_1} e^{A_0(t_1-\tau)} B_0 d\tau \quad (3.2)$$

$$x(t_{r_1}) = e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0}) + \int_{t_{r_0}}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \quad (3.3)$$

subtracting 3.2 and 3.3 we get

$$\begin{aligned} x(t_1) - x(t_{r_1}) &= \overbrace{e^{A_0(t_1-t_0)}x(t_0) - e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0})}^{\square} \\ &\quad + \underbrace{\int_{t_0}^{t_1} e^{A_0(t_1-\tau)} B_0 d\tau - \int_{t_{r_0}}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau}_{\bigcirc} \end{aligned} \quad (3.4)$$

expanding $\square$

$$\begin{aligned} \square &= e^{A_0(t_1-t_0)}x(t_0) - e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0}) \\ &= e^{A_0(t_{r_1}+\delta t_1-t_0)}x(t_0) - e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0}) \\ &= e^{A_0(t_{r_1}-t_0)}e^{\delta t_1}x(t_0) - e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0}) \\ &= F_{r_0}e^{\delta t_1}x(t_0) - F_{r_0}x(t_{r_0}) \\ &= F_{r_0}(I + A_0\delta t_1)x(t_0) - F_{r_0}x(t_{r_0}) \\ &= F_{r_0}[x(t_0) - x_r(t_0)] + F_{r_0}A_0\delta t_1 \end{aligned} \quad (3.5)$$

expanding $\bigcirc$

$$\begin{aligned}
 \bigcirc &= \int_{t_0}^{t_1} e^{A_0(t_1-\tau)} B_0 d\tau - \int_{t_{r_0}}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \\
 &= \int_{t_0}^{t_{r_1}+\delta t_1} e^{A_0(t_{r_1}+\delta t_1-\tau)} B_0 d\tau - \int_{t_0}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \\
 &= e^{A_0\delta t_1} \left[\int_{t_0}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau + \int_{t_{r_1}}^{t_{r_1}+\delta t_1} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \right] - \int_{t_0}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \\
 &= e^{A_0\delta t_1} [G_{r_0} + B_0\delta t_1] - G_{r_0} \\
 &= (I + A_0\delta t_1)[G_{r_0} + B_0\delta t_1] - G_{r_0} \\
 &= \cancel{G_{r_0}} + B_0\delta t_1 + A_0G_{r_0}\delta t_1 + \cancel{A_0B_0\delta t_1\delta t_1} - \cancel{G_{r_0}} \\
 &= A_0G_{r_0}\delta t_1 + B_0\delta t_1
 \end{aligned} \tag{3.6}$$

and replacing (3.5) and (3.6) into equation 3.4 we get to a equation that computes $x(t_1) - x_r(t_{r_1})$ from $x(t_0) - x_r(t_{r_0})$

$$x(t_1) - x_r(t_{r_1}) = F_{r_0}[x(t_0) - x_r(t_{r_0})] + [F_{r_0}A_0x(t_0) + A_0G_{r_0} + B_0]\delta t_1 \tag{3.7}$$

Repeating the proceadure for $x(t_2)$ we have

$$x(t_2) = e^{A_1(t_2-t_1)}x(t_1) + \int_{t_1}^{t_2} e^{A_1(t_2-\tau)} B_1 d\tau \tag{3.8}$$

$$x(t_{r_2}) = e^{A_1(t_{r_2}-t_{r_1})}x(t_{r_1}) + \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)} B_1 d\tau \tag{3.9}$$

subtracting 3.8 and 3.9

$$\begin{aligned}
 x(t_2) - x(t_{r_2}) &= \overbrace{e^{A_1(t_2-t_1)}x(t_1) - e^{A_1(t_{r_2}-t_{r_1})}x(t_{r_1})}^{\square} \\
 &\quad + \underbrace{\int_{t_1}^{t_2} e^{A_1(t_2-\tau)} B_1 d\tau - \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)} B_1 d\tau}_{\bigcirc}
 \end{aligned} \tag{3.10}$$

expanding $\square$

$$\begin{aligned}
 \square &= e^{A_1(t_{r_2}+\delta t_2-t_{r_1}-\delta t_1)}x(t_1) - e^{A_1(t_{r_2}-t_{r_1})}x_r(t_{r_1}) \\
 &= e^{A_1(t_{r_2}-t_{r_1})}e^{A_1(\delta t_2-\delta t_1)}x(t_1) - e^{A_1(t_{r_2}-t_{r_1})}x_r(t_{r_1}) \\
 &= F_{r_1}[I + A_1(\delta t_2 - \delta t_1)]x(t_1) - F_{r_1}x_r(t_{r_1}) \\
 &= F_{r_1}[x(t_1) - x_r(t_{r_1})] + F_{r_1}A_1x(t_1)\delta t_2 - F_{r_1}A_1x(t_1)\delta t_1
 \end{aligned} \tag{3.11}$$

expanding $\bigcirc$

$$\begin{aligned}
 \bigcirc &= \int_{t_1}^{t_2} e^{A_1(t_2-\tau)}B_1d\tau - \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)}B_1d\tau \\
 &= \int_{t_{r_1}+\delta t_1}^{t_{r_2}+\delta t_2} e^{A_1(t_{r_2}+\delta t_2-\tau)}B_1d\tau - \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)}B_1d\tau \\
 &= e^{A_1\delta t_2} \left[\int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)}B_1d\tau - \int_{t_{r_1}}^{t_{r_1}+\delta t_1} e^{A_1(t_{r_2}-\tau)}B_1d\tau + \int_{t_{r_2}}^{t_{r_2}+\delta t_2} e^{A_1(t_{r_2}-\tau)}B_1d\tau \right] \\
 &\quad - \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)}B_1d\tau \\
 &= (I + A_1\delta t_2) \left[G_{r_1} - \int_{t_{r_1}}^{t_{r_1}+\delta t_1} e^{A_1(t_{r_2}-\tau)}B_1d\tau + \int_{t_{r_2}}^{t_{r_2}+\delta t_2} e^{A_1(t_{r_2}-\tau)}B_1d\tau \right] - G_{r_1} \\
 &= (I + A_1\delta t_2) \left[G_{r_1} - e^{A_1(t_{r_2}-t_{r_1})}B_1\delta t_1 + B_1\delta t_2 \right] - G_{r_1} \\
 &= (I + A_1\delta t_2) \left[G_{r_1} - F_{r_1}B_1\delta t_1 + B_1\delta t_2 \right] - G_{r_1} \\
 &= \cancel{G_{r_1}} - F_{r_1}B_1\delta t_1 + B_1\delta t_2 + A_1G_{r_1}\delta t_2 - \cancel{G_{r_1}} \\
 &= A_1G_{r_1}\delta t_2 + B_1\delta t_2 - F_{r_1}B_1\delta t_1
 \end{aligned} \tag{3.12}$$

replacing (3.11) and (3.12) into equation (3.10)

$$\begin{aligned}
 x(t_2) - x_r(t_{r_2}) &= F_{r_1}[x(t_1) - x_{r_1}(t_{r_1})] \\
 &\quad - F_{r_1}[A_1x(t_1) + B_1]\delta t_1 \\
 &\quad + [F_{r_1}A_1 \underbrace{x(t_1)}_{\triangle} + A_1G_{r_1} + B_1]\delta t_2
 \end{aligned} \tag{3.13}$$

expanding $\triangle$ from 3.7

$$\triangle = x(t_1) = x_r(t_{r_1}) + F_{r_0}[x(t_0) - x_r(t_0)] + [F_{r_0}A_0x(t_0) + A_0G_{r_0} + B_0]\delta t_1$$

note that when $\triangle$ is replaced in eq. 3.13 it will produce a product with δt_1 and δt_2 like

$$x(t_1)\delta t_1 = x_r(t_{r_1})\delta t_1 + F_{r_0} \overbrace{[x(t_0) - x_r(t_0)]\delta t_1}^{\square} + [F_{r_0}A_0x(t_0) + A_0G_{r_0} + B_0] \overbrace{\delta t_1\delta t_1}^{\circ}$$

both $\square$ and $\circ$ are products of differences that produces a smaller number and they will computed as ≈ 0

$$x(t_1)\delta t_1 \approx x_r(t_{r_1})\delta t_1 \quad (3.14)$$

replacing equation 3.7 and 3.14 in 3.13

$$\begin{aligned} x(t_2) - x_r(t_{r_2}) = & F_{r_1} \{ F_{r_0} [x(t_0) - x_r(t_0) + [F_{r_0}A_0x(t_0) + A_0G_{r_0} + B_0]\delta t_1] \} \\ & - F_{r_1} [A_1x_r(t_1) + B_1]\delta t_1 \\ & + [F_{r_1}A_1x_r(t_1) + A_1G_{r_1} + B_1]\delta t_2 \end{aligned}$$

$$\begin{aligned} x(t_2) - x_r(t_{r_2}) = & F_{r_1}F_{r_0}[x(t_0) - x_r(t_0)] \\ & + F_{r_1}[F_{r_0}A_0x(t_0) + A_0G_{r_0} - A_1x_r(t_{r_1}) + B_0 - B_1]\delta t_1 \\ & + [F_{r_1}A_1x_r(t_{r_1}) + A_1G_{r_1} + B_1]\delta t_2 \end{aligned} \quad (3.15)$$

Now, repeating the proceadure for $x(t_3)$ we have

$$x(t_3) = e^{A_2(t_3-t_2)}x(t_2) + \int_{t_2}^{t_3} e^{A_2(t_3-\tau)}B_2d\tau \quad (3.16)$$

$$x(t_{r_3}) = e^{A_2(t_{r_3}-t_{r_2})}x(t_{r_2}) + \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau \quad (3.17)$$

subtracting 3.16 and 3.17

$$\begin{aligned} x(t_3) - x(t_{r_3}) = & \overbrace{e^{A_2(t_3-t_2)}x(t_2) - e^{A_2(t_{r_3}-t_{r_2})}x(t_{r_2})}^{\square} \\ & + \underbrace{\int_{t_2}^{t_3} e^{A_2(t_3-\tau)}B_2d\tau - \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau}_{\circ} \end{aligned} \quad (3.18)$$

expanding $\square$

$$\begin{aligned}
 \square &= e^{A_2(t_{r_3} + \delta t_3 - t_{r_2} - \delta t_2)} x(t_2) - e^{A_2(t_{r_3} - t_{r_2})} x_r(t_{r_2}) \\
 &= e^{A_2(t_{r_3} - t_{r_2})} e^{A_2(\delta t_3 - \delta t_2)} x(t_2) - e^{A_2(t_{r_3} - t_{r_2})} x_r(t_{r_2}) \\
 &= F_{r_2} [I + A_2(\delta t_3 - \delta t_2)] x(t_2) - F_{r_2} x_r(t_{r_2}) \\
 &= F_{r_2} [x(t_2) - x_r(t_{r_2})] + F_{r_2} A_2 x(t_2) \delta t_3 - F_{r_2} A_2 x(t_2) \delta t_2
 \end{aligned} \tag{3.19}$$

expanding $\bigcirc$

$$\begin{aligned}
 \bigcirc &= \int_{t_2}^{t_3} e^{A_2(t_3 - \tau)} B_2 d\tau - \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3} - \tau)} B_2 d\tau \\
 &= \int_{t_{r_2} + \delta t_2}^{t_{r_3} + \delta t_3} e^{A_2(t_{r_3} + \delta t_3 - \tau)} B_2 d\tau - \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3} - \tau)} B_2 d\tau \\
 &= e^{A_2 \delta t_3} \left[\int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3} - \tau)} B_2 d\tau - \int_{t_{r_2}}^{t_{r_2} + \delta t_2} e^{A_2(t_{r_3} - \tau)} B_2 d\tau + \int_{t_{r_3}}^{t_{r_3} + \delta t_3} e^{A_2(t_{r_3} - \tau)} B_2 d\tau \right] \\
 &\quad - \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3} - \tau)} B_2 d\tau \\
 &= (I + A_2 \delta t_3) \left[G_{r_2} - \int_{t_{r_2}}^{t_{r_2} + \delta t_2} e^{A_2(t_{r_3} - \tau)} B_2 d\tau + \int_{t_{r_3}}^{t_{r_3} + \delta t_3} e^{A_2(t_{r_3} - \tau)} B_2 d\tau \right] - G_{r_2} \\
 &= (I + A_2 \delta t_3) \left[G_{r_2} - e^{A_2(t_{r_3} - t_{r_2})} B_2 \delta t_2 + B_2 \delta t_3 \right] - G_{r_2} \\
 &= (I + A_2 \delta t_3) \left[G_{r_2} - F_{r_2} B_2 \delta t_2 + B_2 \delta t_3 \right] - G_{r_2} \\
 &= \cancel{G_{r_2}} - F_{r_2} B_2 \delta t_2 + B_2 \delta t_3 + A_2 G_{r_2} \delta t_3 - \cancel{G_{r_2}} \\
 &= A_2 G_{r_2} \delta t_3 + B_2 \delta t_3 - F_{r_2} B_2 \delta t_2
 \end{aligned} \tag{3.20}$$

replacing (3.19) in (3.20) back in eq. (3.18)

$$\begin{aligned}
 x(t_3) - x_r(t_{r_3}) &= F_{r_2} [x(t_2) - x_{r_2}(t_{r_2})] \\
 &\quad - F_{r_2} [A_2 x(t_2) + B_2] \delta t_2 \\
 &\quad + [F_{r_2} A_2 x(t_2) + A_2 G_{r_2} + B_2] \delta t_3
 \end{aligned}$$

$$\begin{aligned}
 x(t_3) - x_r(t_{r_3}) = & F_{r_2} \{ F_{r_1} F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_1} [F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1] \delta t_1 \\
 & + [F_{r_1} A_1 x_r(t_1) + A_1 G_{r_1} + B_1] \delta t_2 \} \\
 & - F_{r_2} [A_2 x_r(t_2) + B_2] \delta t_2 \\
 & + [F_{r_2} A_2 x_r(t_2) + A_2 G_{r_2} + B_2] \delta t_3
 \end{aligned}$$

$$\begin{aligned}
 x(t_3) - x_r(t_{r_3}) = & F_{r_2} F_{r_1} F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_2} F_{r_1} [F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1] \delta t_1 \\
 & + F_{r_2} [F_{r_1} A_1 x_r(t_1) + A_1 G_{r_1} - A_2 x_r(t_{r_2}) + B_1 - B_2] \delta t_2 \\
 & + [F_{r_2} A_2 x_r(t_2) + A_2 G_{r_2} + B_2] \delta t_3
 \end{aligned} \tag{3.21}$$

Continuing on expanding the propagation of the system further more we can deduce the equation for $x(t_i)$ as

$$\begin{aligned}
 x(t_i) - x_r(t_{r_i}) = & F_{r_{i-1}} F_{r_{i-2}} \dots F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_{i-1}} F_{r_{i-2}} \dots F_{r_1} [F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1] \delta t_1 \\
 & + F_{r_{i-1}} F_{r_{i-2}} \dots F_{r_2} [F_{r_1} A_1 x_r(t_1) + A_1 G_{r_1} - A_2 x_r(t_{r_2}) + B_1 - B_2] \delta t_2 \\
 & \vdots \\
 & + F_{r_{i-1}} [F_{r_{i-2}} A_{i-2} x_r(t_{i-2}) + A_{i-2} G_{r_{i-2}} - A_{i-1} x_r(t_{r_{i-1}}) + B_{i-2} - B_{i-1}] \delta t_{i-1} \\
 & + [F_{r_{i-1}} A_{i-1} x_r(t_{i-1}) + A_{i-1} G_{r_{i-1}} + B_{i-1}] \delta t_i
 \end{aligned} \tag{3.22}$$

and then, for $x(t_N)$ we have

$$\begin{aligned}
 x(t_N) - x_r(t_{r_N}) = & F_{r_{N-1}} F_{r_{N-2}} \cdots F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_{N-1}} F_{r_{N-2}} \cdots F_{r_1} \overbrace{[F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1]}^{\square} \delta t_1 \\
 & + F_{r_{N-1}} F_{r_{N-2}} \cdots F_{r_2} [F_{r_1} A_1 x(t_1) + A_1 G_{r_1} - A_2 x_r(t_{r_2}) + B_1 - B_2] \delta t_2 \\
 & \vdots \\
 & + F_{r_{N-1}} [F_{r_{N-2}} A_{N-2} x_r(t_{N-2}) + A_{N-2} G_{r_{N-2}} - A_{N-1} x_r(t_{r_{N-1}}) \\
 & \quad + B_{N-2} - B_{N-1}] \delta t_{N-1} \\
 & + \underbrace{[F_{r_{N-1}} A_{N-1} x_r(t_{N-1}) + A_{N-1} G_{r_{N-1}} + B_{N-1}]}_{\bigcirc} \delta t_N
 \end{aligned} \tag{3.23}$$

$$\begin{aligned}
 \square = & F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1 \\
 = & A_0 F_{r_0} x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1 \\
 = & A_0 [x_r(t_{r_1}) - G_{r_0}] + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1 \\
 = & A_0 x_r(t_{r_1}) - \cancel{A_0 G_{r_0}} + \cancel{A_0 G_{r_0}} - A_1 x_r(t_{r_1}) + B_0 - B_1 \\
 = & (A_0 - A_1) x_r(t_{r_1}) + B_0 - B_1
 \end{aligned} \tag{3.24}$$

$$\begin{aligned}
 \bigcirc = & F_{r_{N-1}} A_{N-1} x_r(t_{N-1}) + A_{N-1} G_{r_{N-1}} + B_{N-1} \\
 = & A_{N-1} F_{r_{N-1}} x_r(t_{N-1}) + A_{N-1} G_{r_{N-1}} + B_{N-1} \\
 = & A_{N-1} [x_r(t_{r_N}) - G_{r_{N-1}}] + A_{N-1} G_{r_{N-1}} + B_{N-1} \\
 = & A_{N-1} x_r(t_{r_N}) - \cancel{A_{N-1} G_{r_{N-1}}} + \cancel{A_{N-1} G_{r_{N-1}}} + B_{N-1} \\
 = & A_{N-1} x_r(t_{r_N}) + B_{N-1}
 \end{aligned} \tag{3.25}$$

replacing 3.24 and 3.25 in 3.23

$$\begin{aligned}
 x(t_N) - x_r(t_{r_N}) = & F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_1} [(A_0 - A_1)x_r(t_{r_1}) + B_0 - B_1] \delta t_1 \\
 & + F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_2} [(A_1 - A_2)x_r(t_{r_2}) + B_1 - B_2] \delta t_2 \\
 & \vdots \\
 & + F_{r_{N-1}} [(A_{N-2} - A_{N-1})x_r(t_{r_{N-1}}) + B_{N-2} - B_{N-1}] \delta t_{N-1} \\
 & + [A_{N-1}x_r(t_{r_N}) + B_{N-1}] \delta t_N
 \end{aligned} \tag{3.26}$$

$$e(t) = x(t) - x_r(t_r) \tag{3.27}$$

$$\Phi = F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_0} \tag{3.28}$$

$$\begin{aligned}
 \Gamma = & \begin{bmatrix} F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_1} [(A_0 - A_1)x_r(t_{r_1}) + B_0 - B_1], \\
 F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_2} [(A_1 - A_2)x_r(t_{r_2}) + B_1 - B_2], \\
 \vdots \\
 F_{r_{N-1}} [(A_{N-2} - A_{N-1})x_r(t_{r_{N-1}}) + B_{N-2} - B_{N-1}] \end{bmatrix}
 \end{aligned} \tag{3.29}$$

$$\delta t = [\delta t_1, \delta t_2, \dots, \delta t_N]^T \tag{3.30}$$

$$e(t_N) = \Phi e(t_0) + \Gamma \delta t \tag{3.31}$$

where x_r is calculated as

$$x_r(t_{r_1}) = F_0 x_r(t_{r_0}) + G_0 \tag{3.32}$$

$$x_r(t_{r_2}) = F_1 x_r(t_{r_1}) + G_1 \tag{3.33}$$

$\vdots$

$$x_r(t_{r_N}) = F_{N-1} x_r(t_{r_{N-1}}) + G_{N-1} \tag{3.34}$$

and F_i and G_i are calculated as

$$[F_i, G_i] = c2dm(Ai, Bi, [], [], dti, 'zoh') \tag{3.35}$$

3.3 Optimal control problem

3.3.1 Cost function

3.3.2 Prediction equation

3.3.3 Constraints

3.3.4 Feasibility region

3.4 Application Examples

3.4.1 Integrador duplo

3.4.2 Patino 1

3.4.3 Patino 2

Matrizes do artigo patino exemplo 2:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & -\frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\frac{R}{L}x_3(t) \end{bmatrix} \quad (3.36)$$

$$A = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & \frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \quad (3.37)$$

$$\begin{aligned}
 \mathbf{u}_0 &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T \\
 \mathbf{u}_1 &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T \\
 \mathbf{u}_2 &= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T \\
 \mathbf{u}_3 &= \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}^T \\
 \mathbf{u}_4 &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T \\
 \mathbf{u}_5 &= \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T \\
 \mathbf{u}_6 &= \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^T \\
 \mathbf{u}_7 &= \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T
 \end{aligned} \tag{3.38}$$

$$A_{u_0} = A\mathbf{u}_0 = \begin{bmatrix} 0 \\ 0 \\ \frac{-Rx_3(t)}{L} \end{bmatrix} \tag{3.39}$$

$$A_{u_1} = A\mathbf{u}_1 = \begin{bmatrix} 0 \\ \frac{x_3(t)}{C_2} \\ \frac{E-x_2(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.40}$$

$$A_{u_2} = A\mathbf{u}_2 = \begin{bmatrix} \frac{x_3(t)}{C_1} \\ \frac{-x_3(t)}{C_2} \\ \frac{x_2(t)-x_1(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.41}$$

$$A_{u_3} = A\mathbf{u}_3 = \begin{bmatrix} \frac{x_3(t)}{C_1} \\ 0 \\ \frac{E-x_1(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.42}$$

$$A_{u_4} = A\mathbf{u}_4 = \begin{bmatrix} \frac{-x_3(t)}{C_1} \\ 0 \\ \frac{x_1(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.43}$$

$$A_{u_5} = A\mathbf{u}_5 = \begin{bmatrix} \frac{-x_3(t)}{C_1} \\ \frac{x_3(t)}{C_2} \\ \frac{E-x_1(t)-x_2(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.44}$$

$$A_{u_6} = A\mathbf{u}_6 = \begin{bmatrix} 0 \\ \frac{-x_3(t)}{C_2} \\ \frac{x_2(t) - Rx_3(t)}{L} \end{bmatrix} \quad (3.45)$$

$$A_{u_7} = A\mathbf{u}_7 = \begin{bmatrix} 0 \\ 0 \\ \frac{E - Rx_3(t)}{L} \end{bmatrix} \quad (3.46)$$

$$\dot{X} = A_{u_0}X + B_0u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (3.47)$$

$$\dot{X} = A_{u_1}X + B_1u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_2} \\ 0 & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (3.48)$$

$$\dot{X} = A_{u_2}X + B_2u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & -\frac{1}{C_2} \\ -\frac{1}{L} & \frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (3.49)$$

$$\dot{X} = A_{u_3}X + B_3u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ -\frac{1}{L} & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (3.50)$$

$$\dot{X} = A_{u_4}X + B_4u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & 0 \\ -\frac{1}{L} & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (3.51)$$

$$\dot{X} = A_{u_5}X + B_5u = \begin{bmatrix} 0 & 0 & -\frac{1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ \frac{1}{L} & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (3.52)$$

$$\dot{X} = A_{u_6}X + B_6u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_2} \\ 0 & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (3.53)$$

$$\dot{X} = A_{u_7}X + B_7u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (3.54)$$

4 Conclusão

Concluimos que esse eh o capitulo 4...

Bibliography

Appendix A - Tópicos de Dilema Linear

A.1 Modelo de simulação

```
+engine
├─ CORES.m
├─ fun_custo_patino.m
├─ get_config_sim_matheus.m
├─ get_config_sim_patino_1.m
├─ get_config_sim_patino_2.m
├─ get_config.m
├─ get_otmin_opt.m
├─ get_ts.m
├─ get_x0.m
├─ get_xr.m
├─ nonlincon_patino.m
├─ otmin.m
├─ sim_1.m
├─ sim_n.m
└─ +mpc
    ├─ construcao_modelo_instantes.m
    ├─ matrizes_ss_mpc_dualmode_switching.m
    └─ mpc_dualmode_switching.m
```

A.1.1 get_x0

funcao de calculo da condicao inicial

A.1.2 get_xf

Texto do modelo de simulação sendo colocado aqui.

Annex A - Exemplo de um Primeiro Anexo

A.1 Uma Seção do Primeiro Anexo

Algum texto na primeira seção do primeiro anexo.

FOLHA DE REGISTRO DO DOCUMENTO

1. CLASSIFICAÇÃO/TIPO TD	2. DATA 25 de março de 2015	3. DOCUMENTO Nº DCTA/ITA/DM-018/2015	4. Nº DE PÁGINAS 35
5. TÍTULO E SUBTÍTULO: Robust control of linear systems with Switched Actuators Subjected to Dwell-Time Constraints			
6. AUTOR(ES): Daniel Vieira			
7. INSTITUIÇÃO(ÕES)/ÓRGÃO(S) INTERNO(S)/DIVISÃO(ÕES): Instituto Tecnológico de Aeronáutica – ITA			
8. PALAVRAS-CHAVE SUGERIDAS PELO AUTOR: Cupim; Cimento; Estruturas			
9. PALAVRAS-CHAVE RESULTANTES DE INDEXAÇÃO: Cupim; Dilema; Construção			
10. APRESENTAÇÃO: (X) Nacional () Internacional ITA, São José dos Campos. Curso de Mestrado. Programa de Pós-Graduação em Engenharia Aeronáutica e Mecânica. Área de Sistemas Aeroespaciais e Mecatrônica. Orientador: Prof. Dr. Adalberto Santos Dupont. Coorientadora: Prof ^a . Dr ^a . Doralice Serra. Defesa em 05/03/2015. Publicada em 25/03/2015.			
11. RESUMO: Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o systema à trajetória cíclica alvo.			
12. GRAU DE SIGILO: (X) OSTENSIVO () RESERVADO () SECRETO			